Introduction to Programming:

DECISION MAKING AND BRANCHING

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Outline

- Decision Making in C Programming
- If Statement
- Nested If Statement
- Else .. If Ladder
- Switch Statement
- Conditional Operator Statement
- Goto Statement
- Decision Making with Logical Operators

Decision Making

- C program statements are executed sequentially.
- Decision Making statements are used to control the flow of program.
- It allows us to control whether a program segment is executed or not.
- It evaluates condition or logical expression first and based on its result (either true or false), the control is transferred to particular statement.
- If result is true then it takes one path else it takes another path.

Decision Making

Decision Making Statements are:

- One way Decision: if (Also known as simple if)
- Two way Decision: if…else
- Multi way Decision: if…else if…else if…else
- Two way Decision: ?: (Conditional Operator)
- n-way Decision: switch…case



Relational Operator

- Relational Operator is used to compare two expressions
- It gives result either true or false based on relationship of two expressions.

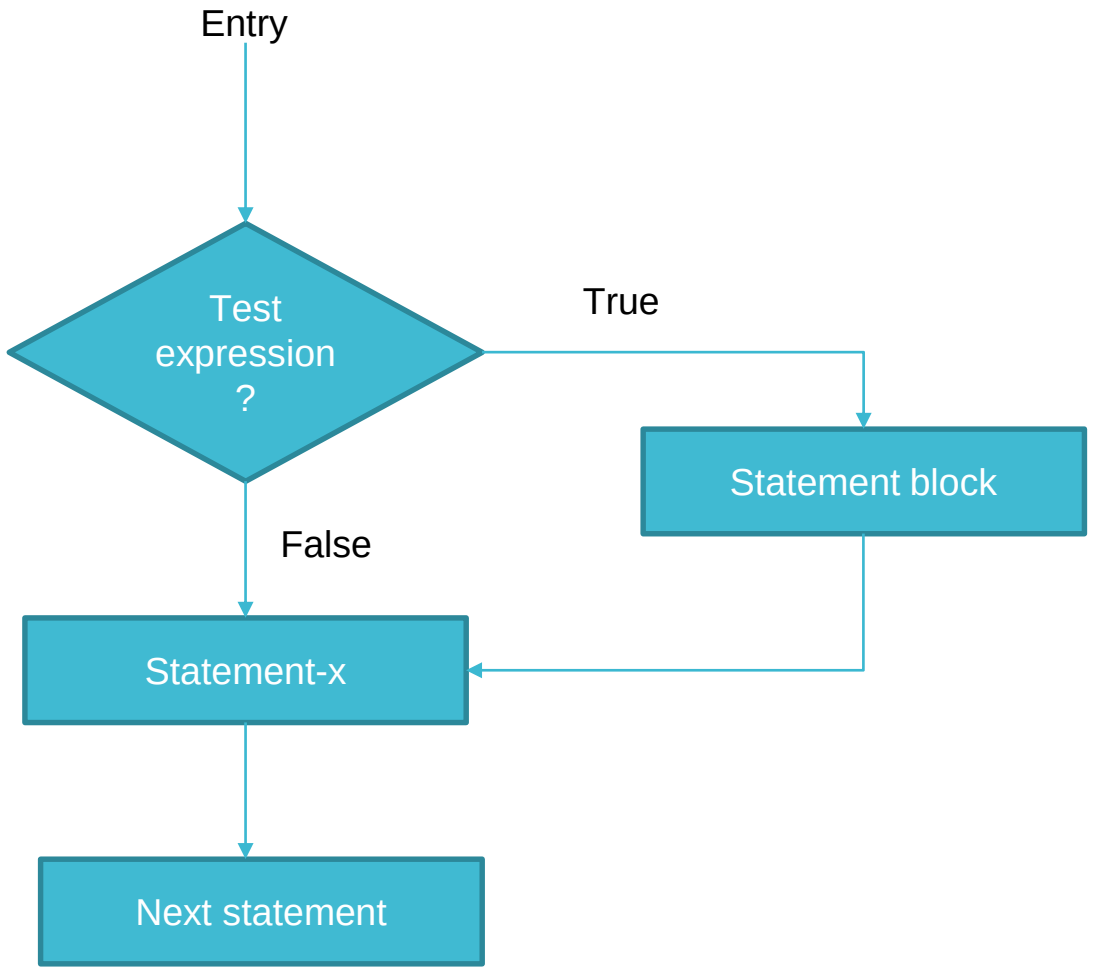
if statement

- if is single branch decision making statement.
- If condition is true then only body will be executed.
- if is a keyword.

Syntax:

```
if(test expression)
{
        Statement-block;
}
statement-x;
```

if statement



If Example:

```
#include<stdio.h>
void main()

{

        int a;

        printf("Enter Number:");

        scanf("%d",&a);

        if(a == 0)

        {

                printf("Zero");

        }

}
```

O/p:
Enter no: 0
Zero

If Example:

WAP to print Positive or Negative Number

```
#include<stdio.h>
void main()
{
        int a;

        printf("Enter Number:");
        scanf("%d",&a);

        if(a >= 0)
        {
                printf("Positive Number");
        }
        if(a < 0)
        {
                printf("Negative Number");
        }

}
```

If Example

WAP to print Odd or Even Number

```
#include<stdio.h>
void main()
{
        int a;

        printf("Enter Number:");
        scanf("%d",&a);

        if(a%2 == 0)
        {
                printf("Even Number");
        }
        if(a%2 != 0)
        {
                printf("Odd Number");
        }
}
```

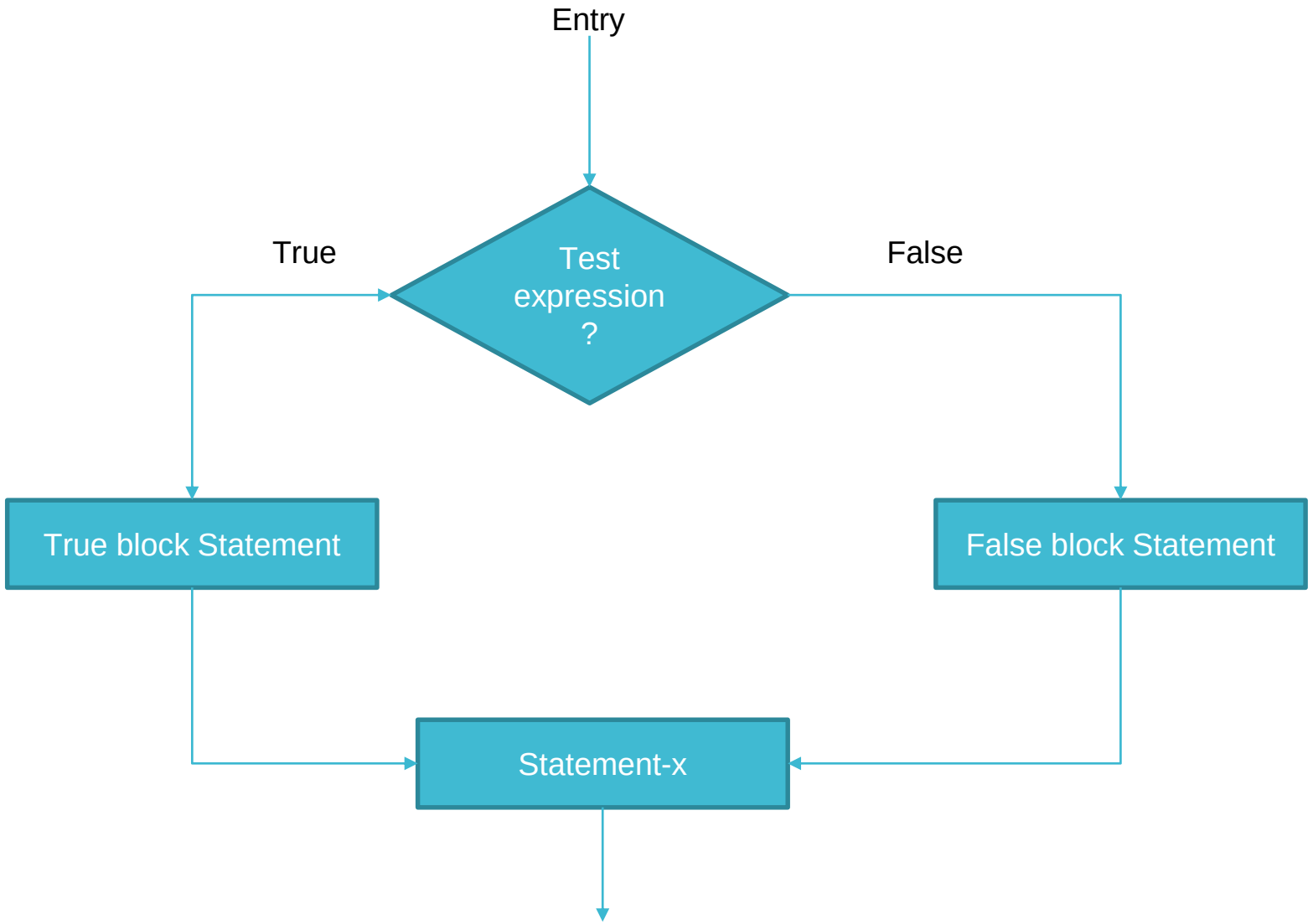
If…else statement

- if…else is two branch decision making statement
- If condition is true then true part will be executed else false part will be executed
- else is keyword
- Syntax:

```
if(condition)
{
        // true part
}
else
{
        // false part
}
```



If…else statement



If...else Example:

Write a program to print Positive or Negative Number

```
#include<stdio.h>
void main()
{
        int a;

        printf("Enter Number:");
        scanf("%d",&a);

        if(a >= 0)
        {
                printf("Positive Number");
        }
        else
        {
                printf("Negative Number");
        }

}
```

If...else Example

Write a program to print Odd or Even Number

```
#include<stdio.h>
void main()
{
        int a;

        printf("Enter Number:");
        scanf("%d",&a);

        if(a%2 == 0)
        {
                printf("Even Number");
        }
        else
        {
                printf("Odd Number");
        }
}
```

If…else Example

Write a program to find maximum number from given 2 numbers

```
#include<stdio.h>
void main()
{
        int a,b;

        printf("Enter Number:");
        scanf("%d",&a);
        printf("Enter Number:");
        scanf("%d",&b);

        if(a>b)
        {
                printf(“%d is maximum“,a);
        }
        else
        {
                printf(“%d is maximum“,b);

        }
}
```

else if Ladder

- if…else if…else if…else is multi branch decision making statement.
- If first if condition is true then remaining if conditions will not be evaluated.
- If first if condition is false then second if condition will be evaluated and if it is true then remaining if conditions will not be evaluated.
- if…else if…else if…else is also known as if…else if ladder

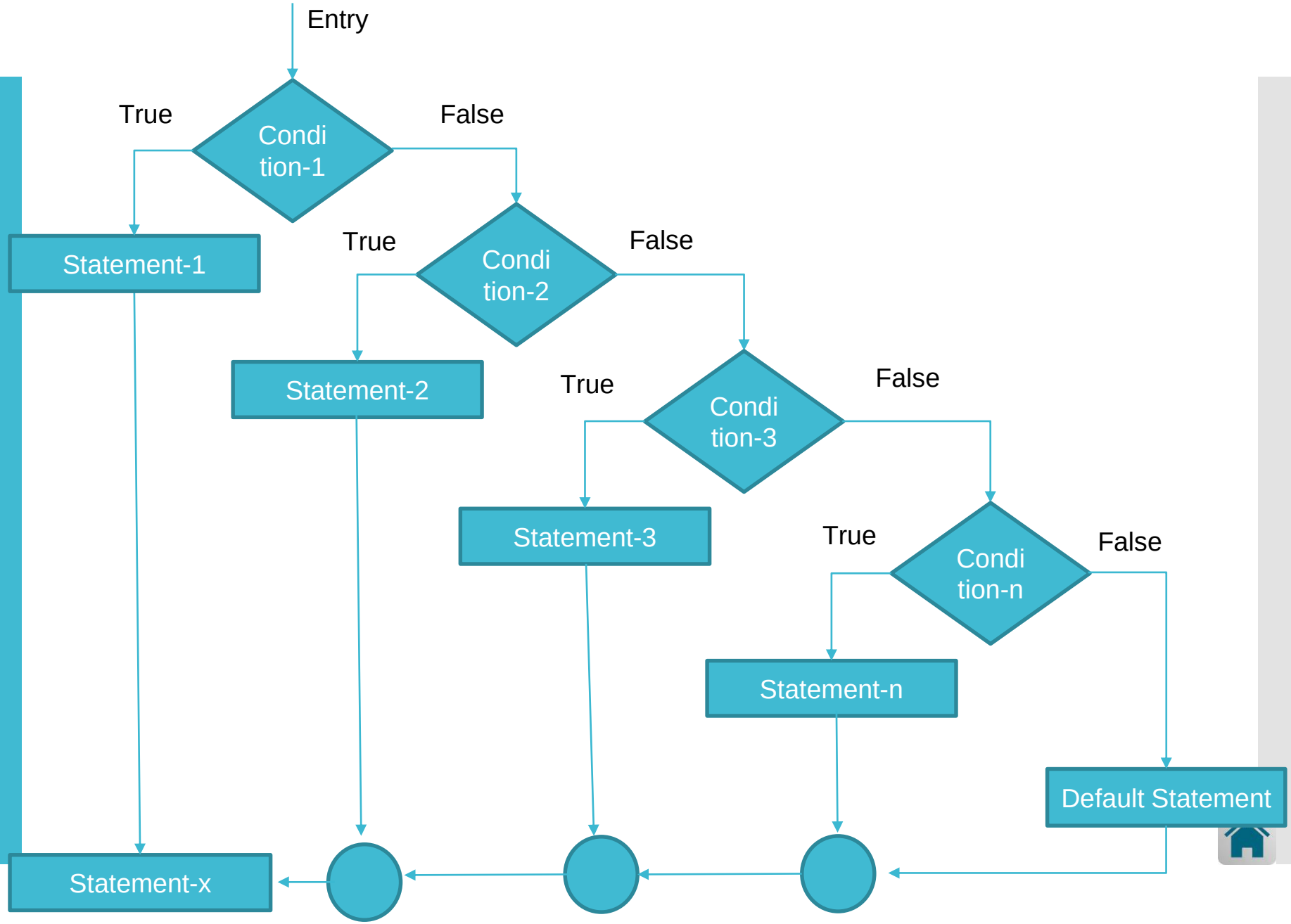
Syntax:

```
if(condition-1)
    statement-1;
elseif(condition-2)
            statement-2;
    elseif(condition-3)
            statement-3;

    else
            default statement;;
```



elseif ladder



else if Ladder Example:

Write a program to print Zero, Positive or Negative Number

```
#include<stdio.h>
void main()
{
        int a;

        printf("Enter Number:");
        scanf("%d",&a);

        if(a > 0)
        {
                printf("Positive Number");
        }
        elseif(a==0)
        {
                printf(“Zero");
        }
        else
        {
                printf("Negative Number");
        }

}
```

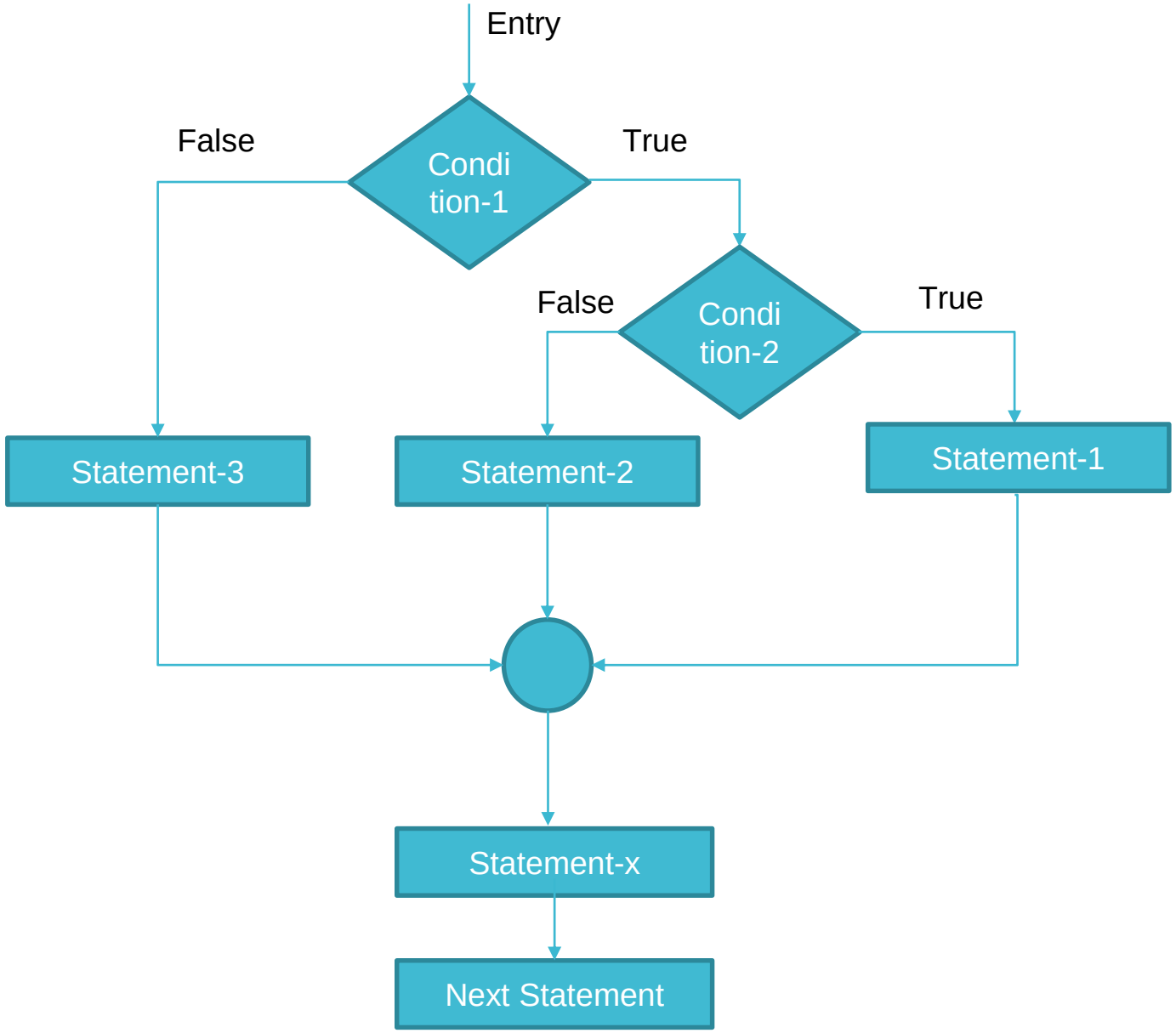
Nested if Statement

- If condition-1 is true then condition-2 is evaluated. If it is true then statement-1 will beexecuted.
- If condition-1 is false then statement-3 will be executed.
- Syntax:

```
if(condition-1)
{
        if(condition-2)
        {
                statement-1;
        }
        else
        {
                statement-2;
        }
}
else
{
        statement-3;
}
```



Nested if ladder



Nested if Example

Write a program to find maximum number from given 3 numbers

```
#include<stdio.h>
void main()
{
        int a, b, c;
        printf("Enter Three Numbers:");
        scanf("%d%d%d",&a,&b,&c);

        if(a>b)
        {
                if(a>c)
                        printf("%d is max",a);
                else
                        printf("%d is max",c);
        }
        else
        {
                if(b>c)
                        printf("%d is max",b);
                else
                        printf("%d is max",c);
        }
}
```

Conditional Operator

- The conditional works operator is similar to the if-else.
- It is also known as a ternary operator.
- It returns first value of expression (before colon(:)) if expression is true and second value of expression if expression is false.
- Syntax:

  variable = Expression1 ? Expression2 : Expression3

  - Here, Expression1 = the condition to be evaluated.
  - Expression2 will be executed and the result will be returned. If the condition (Expression1) is True then
  - Otherwise, if condition(Expression1) is false then Expression3 will be executed and the result will be returned.



Conditional Operator Example

Write a program to find maximum number from given 2 numbers

```
#include<stdio.h>
void main()
{
        int a,b,max;

        printf("Enter Number:");
        scanf("%d",&a);
        printf("Enter Number:");
        scanf("%d",&b);

        max = a>b ? a : b;

        printf("%d is maximum",max);

}
```

Switch..case Statement

- The switch statement allows to execute one code block among many alternatives.
- It works similar to if…else..if ladder.
- Syntax:

```
switch (expression)
{
        case constant1:
                // statements
                break;
        case constant2:
                // statements
                break;
        .
        .
        .
        default:
              // default statement
}
```

Switch..case Statement

- The expression is evaluated once and compared with the values of each case.
- If there is a match, the corresponding statements after the matching case are executed.
- If there is no match, the default statements are executed.
- If we do not use break, all statements after the matching label are executed.
- The default clause inside the switch statement is optional.



Switch..case Example

Write a program that asks week day number and prints week day name

```
#include<stdio.h>
void main()
{
        int day;

        printf("Enter day number ():");-
        scanf("%d",&day);

         switch(day)
        {
                case 1:
                        printf("Sunday");
                        break;
                case 2:
                        printf("Monday");
                        break;
                case 3:
                        printf("Tuesday");
                        break;
                case 4:
                        printf("Wednesday");
                        break;
                case 5:
                        printf("Thursday");
                        break;
```

Switch..case Example

```
                case 6:
                        printf("Friday");
                        break;
                case 7:
                        printf("Saturday");
                        break;
                default:
                        printf("Wrong input");
                        break;
        }
}
```

Output:

```
Enter day number(1-7):5
Thursday
```

#Thank You

From @:Darsha

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